Wall-effect gain chain: $c(\bar{h}) \to a \to a' \to a''$

(z axis, perpendicular; $\bar{h} := h/R$)

Symbol	Value	Unit
R	2.25	μm
γ_N	0.0425	pN·s/μm
Δt	1/1600	s
$a_{nom} = \Delta t/\gamma_N$	1.4706×10^{-2}	μm/pN

$$a' := \frac{da}{dh}, \qquad a'' := \frac{d^2a}{dh^2}, \qquad D := \frac{1}{c}, \qquad K_h := \frac{1}{c}\frac{dc}{d\bar{h}}, \qquad K_h' := \frac{dK_h}{d\bar{h}}$$

1. Wall correction c

$$c(\bar{h}) = \frac{1}{D(\bar{h})}, \qquad D(\bar{h}) = 1 - \frac{9}{8\bar{h}} + \frac{1}{2\bar{h}^3} - \frac{57}{100\,\bar{h}^4} + \frac{1}{5\bar{h}^5} + \frac{7}{200\,\bar{h}^{11}} - \frac{1}{25\,\bar{h}^{12}}$$

2. Gain a

$$\boxed{a(\bar{h}) = \frac{a_{nom}}{c(\bar{h})} = a_{nom}\,D(\bar{h})}$$

$$a(\bar{h}) = a_{nom}\left[1 - \frac{9}{8\bar{h}} + \frac{1}{2\bar{h}^3} - \frac{57}{100\,\bar{h}^4} + \frac{1}{5\bar{h}^5} + \frac{7}{200\,\bar{h}^{11}} - \frac{1}{25\,\bar{h}^{12}}\right]$$

3. Slope a'

$$\boxed{a'(\bar{h}) = -\frac{a\,K_h}{R} = \frac{a_{nom}}{R}\,D'(\bar{h})}$$

$$a'(\bar{h}) = \frac{a_{nom}}{R}\left[\frac{9}{8\bar{h}^2} - \frac{3}{2\bar{h}^4} + \frac{57}{25\,\bar{h}^5} - \frac{1}{\bar{h}^6} - \frac{77}{200\,\bar{h}^{12}} + \frac{12}{25\,\bar{h}^{13}}\right]$$

4. Curvature a''

$$\boxed{a''(\bar{h}) = \frac{a}{R^2}\left(K_h^2 - K_h'\right) = \frac{a_{nom}}{R^2}\, D''(\bar{h})}$$

$$a''(\bar{h}) = \frac{a_{nom}}{R^2}\left[-\frac{9}{4\bar{h}^3} + \frac{6}{\bar{h}^5} - \frac{57}{5\,\bar{h}^6} + \frac{6}{\bar{h}^7} + \frac{231}{50\,\bar{h}^{13}} - \frac{156}{25\,\bar{h}^{14}}\right]$$

5. Curves